

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** ServiceNow Employee Training Records – Import, Dot-Walking and ACL Security
- **Team Members:** Not provided in the supplied project details. The report therefore does not invent team member names or roles.

2. Project Overview

• **Purpose:** The project links each training record to an employee and exposes employee information such as department through dot-walking for easier reporting. It also implements role-based access control using an HR Manager role and Access Control Lists (ACLs), with impersonation used to validate access.

• **Features:**

- Custom Employee Training table with Training Name, Completion Date, Status, and Employee fields.
- Employee reference to the ServiceNow **sys_user** table.
- Spreadsheet import through System Import Sets and Transform Maps.
- Dot-walking from Employee to Department.
- HR Manager role and Read/Write ACLs.
- Impersonation-based testing for authorized and unauthorized users.

3. Architecture

- **Frontend:** The supplied project is a ServiceNow platform configuration. It does not use a separate React frontend. Users interact through the ServiceNow interface, All menu, application modules, forms, lists, and administration pages.
- **Backend:** The project does not use a separate Node.js/Express backend. ServiceNow provides the platform-side processing for tables, import sets, transform maps, roles, ACL evaluation, and user impersonation.
- **Database:** The Employee Training Records table stores Training Name, Completion Date, Status, and an Employee reference to **sys_user**. Dot-walking is used to access the employee's Department information.

ServiceNow Employee Training Records Architecture

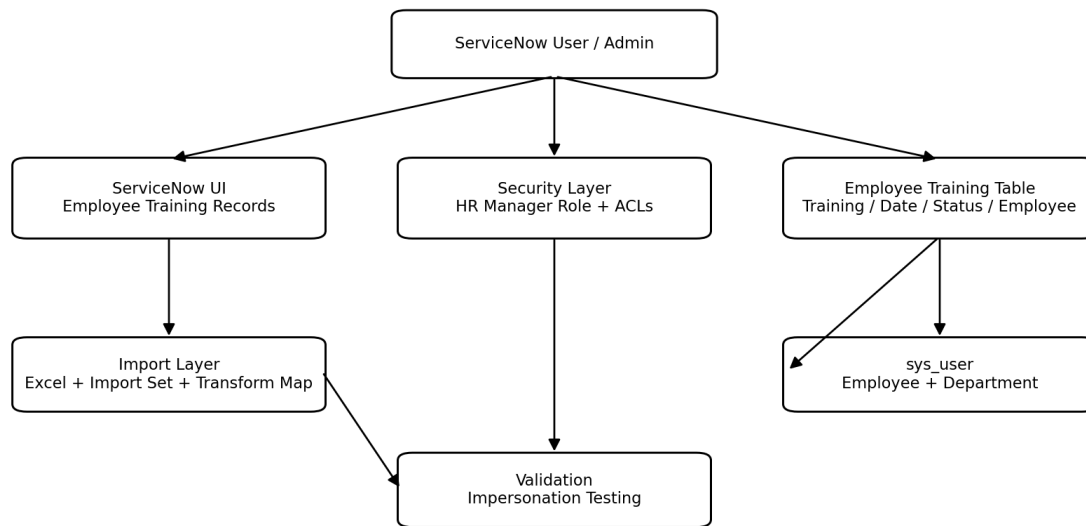


Figure 1. ServiceNow architecture adapted to the supplied project requirements.

4. Setup Instructions

- **Prerequisites:** Access to a ServiceNow instance with permission to create tables, import data, configure roles, and create ACLs. A spreadsheet containing Training Name, Completion Date, Status, and Employee is also required.
- **Installation / Configuration:**
 1. Open ServiceNow and navigate to All → Tables under System Definition.
 2. Create the Employee Training table and add the four specified fields.
 3. Add Status choices through the Dictionary Entry.
 4. Use System Import Sets → Load Data to upload the spreadsheet.
 5. Create a Transform Map and add the required field mappings.
 6. Create the backup user and HR Manager role, then assign the role.
 7. Elevate to security_admin and create Read and Write ACLs for Employee Training Records.

5. Folder Structure

The supplied project is a ServiceNow configuration rather than a conventional MERN application, so the template's Client/Server folder structure does not apply. The equivalent logical structure is:

ServiceNow Project	Main configuration area
Employee Training Records	Custom table / application data
Training Name	String field
Completion Date	Date field
Status	Choice field
Employee	Reference field → sys_user
Import Set	Spreadsheet loading configuration
Transform Map	Source-to-target field mapping
Dot-Walking	Employee → Department
User / Role	backup user + HR Manager
ACL	Read and Write access controls
Impersonation	Access validation

6. Running the Application

The supplied project does not specify local npm commands because it is implemented inside ServiceNow. The operational flow is performed in the ServiceNow instance.

- Open the ServiceNow instance and use the All menu.
- Open Employee Training Records to create, view, and test records.
- Use System Import Sets to load spreadsheet data.
- Use User Administration for the backup user and HR Manager role.
- Use Profile → Elevate Role to select security_admin when configuring ACLs.

7. API Documentation

No REST API endpoints, request methods, parameters, or JSON responses are provided in the supplied project details. The implementation described is based on ServiceNow UI configuration and platform features rather than a documented custom API.

The main platform operations documented by the project are:

Operation	ServiceNow feature	Purpose
Create / configure table	Tables / System Definition	Create Employee Training Records structure
Load records	System Import Sets	Upload spreadsheet data
Map records	Transform Maps	Map source fields to target fields
Access related data	Dot-Walking	Retrieve Department through Employee
Authorize access	Roles + ACLs	Control Read / Write access
Validate access	Impersonation	Test behavior for different users

8. Authentication

The supplied project describes authorization rather than an application-specific login implementation. Access is controlled using ServiceNow users, the HR Manager role, and ACLs. The backup user is assigned the HR Manager role. A security_admin role is temporarily elevated for ACL configuration. Impersonation is then used to validate how access changes between users.

The project does not specify JWT tokens, session code, OAuth configuration, or custom authentication APIs, so these are not added to the report.

9. User Interface

The user interface consists of ServiceNow forms, list views, the All navigation menu, application/module configuration, user administration screens, and ACL configuration screens. The Employee field can be used with dot-walking so that Department information can be shown in the list view.

Project Workflow

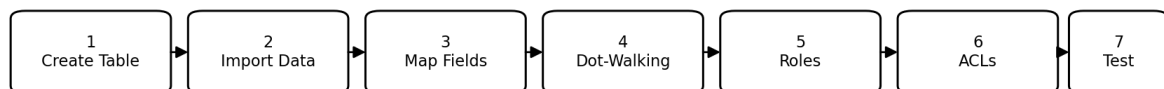


Figure 2. User-facing configuration workflow based on the project milestones.

10. Testing

Testing is based on the result described in the project: impersonate the authorized sys_user and search Employee Training Records; the authorized user can see and edit the permitted fields. When another user is impersonated, the table is not visible.

Impersonation-Based Access Testing

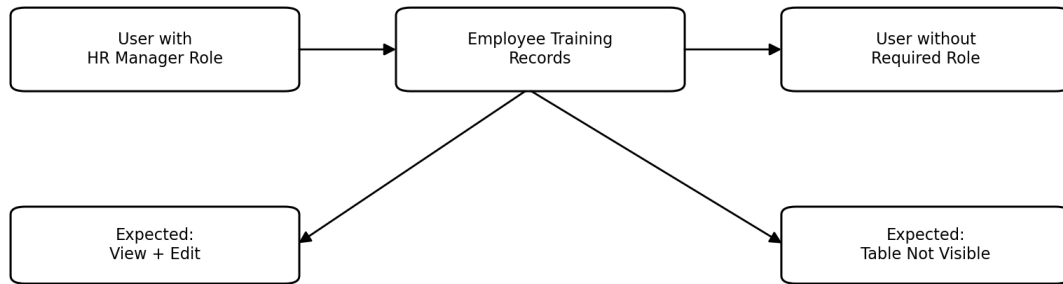


Figure 3. Impersonation-based access testing flow.

ID	Test	Expected Result
TC-01	Open Employee Training Records as authorized user	Table is visible
TC-02	Open/create a training record	Training fields are available
TC-03	Use Employee dot-walking	Department can be shown in the list view
TC-04	Impersonate user with HR Manager role	User can see and edit permitted fields
TC-05	Impersonate another user	Employee Training Records table is not visible
TC-06	Run Transform Map	Imported values are mapped to the target table
TC-07	Read ACL	HR Manager role receives Read access
TC-08	Write ACL	HR Manager role receives Write access

11. Screenshots or Demo

No actual ServiceNow screenshots or demo URL were included in the supplied project details. To avoid presenting fabricated screenshots as real evidence, the report uses process diagrams to represent the configured workflow. The stated testing outcome is documented in Section 10.

Data Flow

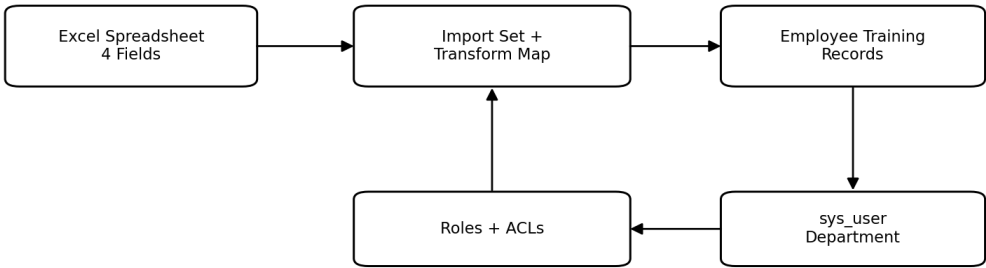


Figure 4. Data-flow illustration based strictly on the fields and ServiceNow features specified in the project.

12. Known Issues

The following limitations are directly identifiable from the supplied material:

- The actual spreadsheet dataset is not included in the provided project description.
- No actual ServiceNow screenshots are supplied in the material used for this report.
- No custom API endpoints or source-code repository are documented.
- The Transform Map field mapping details are described as an activity but the exact mapping table is not provided.
- Access behavior depends on correct role assignment and ACL configuration.

13. Future Enhancements

Potential extensions, consistent with the current solution, include:

- Add automated reminders or notifications for training completion dates.
- Create HR dashboards and reports grouped by employee, department, status, and completion date.
- Add more granular field-level ACLs where required.
- Introduce approval workflows for selected training records.
- Schedule recurring imports for externally maintained training spreadsheets.
- Add additional training attributes and reporting filters.

These are proposed future enhancements; the supplied project does not state that they are already implemented.